



MIKE TRZASKA

WORK EXPERIENCE

ML / AI Engineer

TCC GmbH, Hamburg, Hamburg

October 2025 – Present

Integrated an end-to-end MLOps pipeline for medical data, covering EDA, model training, and deployment. Acted as lead developer for a multimodal deep learning architecture (time-series ECG + scalar features) and LLM agents for medical applications.

Achievements: Successfully trained a ResNet-based model on the newly established MLOps infrastructure and deployed LLM agents.

Master Thesis

Eppendorf SE, Hamburg, Hamburg

April 2025 – September 2025

As part of my thesis, I developed and compared several LLM-based techniques for semantic parameterization of laboratory protocols. Among them, I introduced OLLM (Ontology-guided Large Language Model), which integrates an ontology as external knowledge base. I also built a ground truth dataset and an evaluation framework.

Achievements: The OLLM is accepted to be integrated into automated lab software.

Working Student

Eppendorf Liquidhandling GmbH, Hamburg, Hamburg

October 2022 – September 2025

Responsible for maintaining and extending a GUI testing tool for embedded systems. Involved in solving a variety of software development tasks across different projects. Conducting research on the semantic parameterization of laboratory protocols using LLMs.

Achievements: Contributed to the continuous improvement of internal tooling for embedded testing and conducted early research on the integration of LLMs into laboratory software workflows, which laid the foundation for my subsequent master's thesis.

Bachelor Thesis

Eppendorf Liquidhandling GmbH, Hamburg, Hamburg

April 2022 – September 2022

Developed a model-based testing framework for GUIs in embedded systems, using GUI libraries to automate interactions and validate behaviors. The framework was designed for scalability and maintainability through model abstraction.

Achievements: Laid the foundation for a reusable GUI testing tool that is now actively used in several embedded software projects across the company.

Working Student

Eppendorf Liquidhandling GmbH, Hamburg, Hamburg

April 2021 – March 2022

Developed hardware prototypes by integrating components on STM32 boards. Programmed, extended, and tested embedded software modules.

Achievements: Built functional hardware prototypes for early-stage development and improved modular code design.

Mandatory Internship

Eppendorf SE, Hamburg, Hamburg

February 2021 – March 2021

Enhanced a unit testing tool for embedded systems to execute directly on physical devices. Developed scripts for Git-based CI/CD workflows to enable continuous testing.

CONTACT INFORMATION

Mobile number:
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Website:
<https://hokagem.github.io/>

GitHub:
<https://github.com/HokageM>

LinkedIn:
www.linkedin.com/in/mike-trzaska-b576a6201

E-MAIL:
m.trzaska663@gmail.com

LANGUAGES

German (native)
Englisch (fluent)

TECH STACK

Programming Languages:

C++, Python, TypeScript, C

AI / Machine Learning:

PyTorch, Scikit-learn, HuggingFace, OpenCV, Imbalanced-learn

GenAI & LLM Orchestration:

LangGraph, LangChain, AWS Bedrock, Vertex AI, Langfuse, Boto3, OpenAI

Data Science & Analytics:

NumPy, Pandas, Matplotlib, Jupyter, SciPy

MLOps & DevOps:

MLFlow, Docker, GitHub Actions, PostgreSQL, Git, Terraform

Building Systems & Testing:

UV, PyTest, CMake, Conan, Catch2

Frameworks & Web:

FastAPI, React, Flutter

HOBBIES

Basketball
Calisthenics

Achievements: Enabled reliable test execution on hardware targets and improved CI/CD integration for automated quality assurance.

Internship

Eppendorf Liquidhandling GmbH, Hamburg, Hamburg

October 2020 – January 2021

Developed an emulator-based unit testing framework for embedded systems. Integrated the framework into a GitHub CI/CD workflow within an Agile/Scrum team.

Achievements: Designed a flexible automated testing infrastructure still actively used across multiple projects.

Student Tutor

Technische Universität Hamburg, Hamburg, Hamburg

September 2019 – August 2020

Tutored modules *Object-Oriented Programming and Algorithms & Data Structures* and *Computing in Mechanical Engineering*.

Achievements: Gained experience in teaching and communicating technical concepts to large groups of students.

EDUCATION

Master of Science in Bioinformatics

University of Hamburg

October 2022 – September 2025

Final grade: 1,56

Bachelor of Science in Computer Science and Engineering

Technical University of Hamburg

October 2018 – September 2022

Final grade: 1,7

Abitur, Major: Chemistry

Stadteilschule Poppenbüttel

August 2015 – July 2018

Final grade: 2,3

CERTIFICATIONS

Advanced Machine Learning on Google Cloud

Google Cloud, 2026

Specialization focused on production ML systems, natural language processing, computer vision, and recommendation systems on Google Cloud.

IBM RAG and Agentic AI

IBM, 2026

Specialization in RAG, agentic AI, LangChain, and LangGraph.

Machine Learning Operations (MLOps) with Vertex AI: Manage Features

Google Cloud, 2026

Course covering MLOps tools and best practices for deploying, evaluating, monitoring, and operating production ML systems on Google Cloud.

DATA SCIENCE & MACHINE LEARNING PROJECTS

IRL with Python:

Implemented various inverse reinforcement learning algorithms in Python and PyTorch to solve the MountainCar-v0 task.

Lung Cancer Risk Prediction System:

Built a full-stack web app for lung cancer risk prediction with React, FastAPI, and PostgreSQL. Includes secure doctor authentication, encrypted patient data, and decision-tree-based prediction.

HeartPredict:

Developed a Python library to analyze and predict heart failure outcomes using Scikit-learn, including classification, regression, and survival analysis.